Gradient Descent (Continued)

Gradient :- A vector containing partial derivatives of a function

Partial Derivative :-

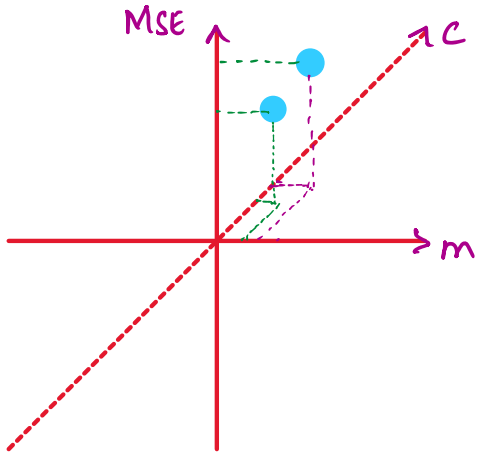
Let us suppose → F ⇝ This function is dependent on multiple parameters, say y, z

∴ change in function F will take place w.r.t. y and z

Similarly, if we apply the same logic to the MSE / Loss function, we can see that it is dependent on both m & c,

∴ Change in loss will take place w.r.t. m & c

which means that if we take the differentiation of loss w.r.t. m & c then this differentiation is partial differentiation of loss



This is for a 2D equation → m, c

Similarly, for n-dimensional data →

eq :- $y = m_1x_1 + m_2x_2 + m_3x_3 + \dots + m_nx_n + c$

↓

Loss is partially changing w.r.t.

$m_1, m_2, m_3, m_4 \dots, m_n, c$

$m_1, m_2, m_3, m_4 \dots, m_n, c$

As we can see that loss is nothing but partial differentiation

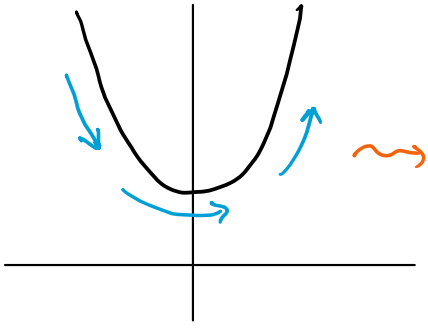
so,

$$\text{Gradient of loss} = (\nabla L) = \begin{bmatrix} \frac{\partial L}{\partial m_1} \\ \frac{\partial L}{\partial m_2} \\ \frac{\partial L}{\partial m_3} \\ \vdots \end{bmatrix}$$

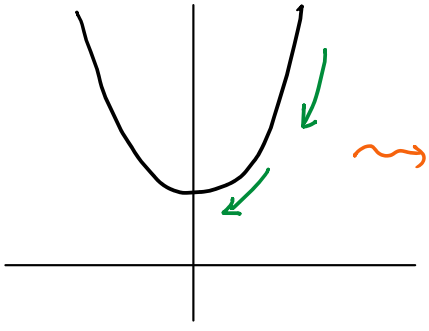
∇ = gradient

$\frac{\partial}{\partial x}$ = partial diff.

Now,



⇝ This is known as moving in the direction of gradient [loss +ve]



⇝ This is known as moving in the opposite direction of gradient [loss -ve]

↓

Decreasing gradient [descent]

Due to this, the algorithm is known as

"Gradient Descent Algorithm"

"Gradient Descent Algorithm"

Final Steps of Linear Regression :-

1. Assume m & c
2. Calculate gradient of loss w.r.t. m & c
3. Update the value of m & c

↓

How to observe that this m & c is perfect m & c

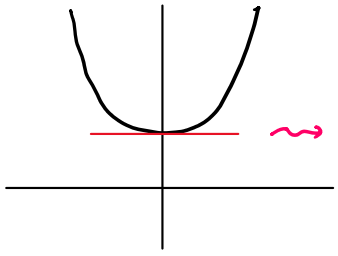
↓

We update it until the value of m & c stops changing

$$m^* = m - \eta \frac{\partial L}{\partial m}$$

$$m^* = m - 0$$

↳ $\frac{\partial L}{\partial m} = 0$, $\frac{\partial L}{\partial c} = 0$



↝ angle 0 or $\frac{\partial L}{\partial m} = 0$

This indicates that we have reached the minimum value (minima) of the loss function

↓

whatever m & c we have obtained here are the perfect values of m & c

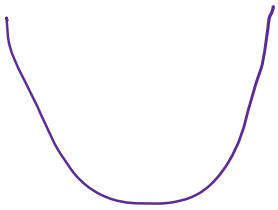
4. Draw a best fitting line using these m & c values

(4) Draw a best fitting line using these m & c values

This completes the model training

(5) Evaluate model's performance

NOTE:-



→ square function is continuous
for differentiation we need continuity
That is why we take square and
not modulus for making errors +ve